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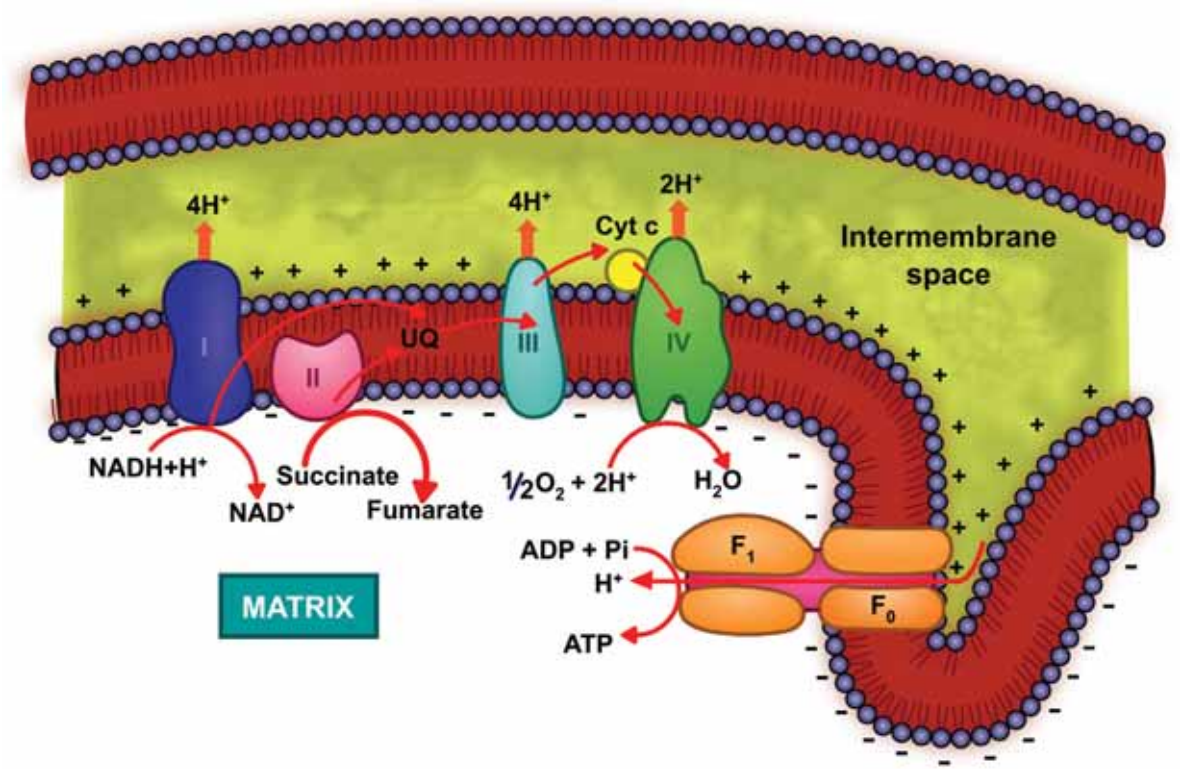


Figure 14.10: Electron Transport Chain and Terminal Oxidation

mitochondrial electron transport chain. However, in those animal tissues in which malate shuttle mechanism is present, the oxidation of external NADH will yield almost 3 ATP molecules.



Abnormal rise in respiratory rate of ripening in fruits is called **Climacteric**. Examples are apple, banana, mango, papaya, pear.

Complete oxidation of a glucose molecule in aerobic respiration results in the net gain of **36 ATP molecules in plants** as shown in table 14.2. Since huge amount of energy is generated in mitochondria in the form of ATP molecules they are called ***'power house of the cell'.*** In the case of aerobic prokaryotes due to lack of mitochondria each molecule of glucose produces 38 ATP molecules.

Recent view

When the cost of transport of ATPs from matrix into the cytosol is considered, the number will be **2.5 ATPs for each NADH + H^+ and 1.5 ATPs for each $FADH_2$** oxidised during electron transport system. Therefore, in plant cells net yield of 30 ATP molecules for complete aerobic oxidation of one molecule of glucose. But in those animal cells (showing malate shuttle mechanism) net yield will be 32 ATP molecules.

Electron transport chain inhibitors

1. **2,4 DNP (Dinitrophenol)** - It prevents synthesis of ATP from ADP, as it directs electrons from Co Q to O_2
2. **Cyanide** - It prevents flow of electrons from Cytochrome a_3 to O_2
3. **Rotenone** - It prevents flow of electrons from NADH + H^+/$FADH_2$ to Co Q
4. **Oligomycin** – It inhibits oxidative phosphorylation